

QTL Mapping of Cotton Fibre Quality Traits: A Comparison of Classical and Regularised Approaches

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1 Introduction

Quantitative trait locus (QTL) mapping aims to identify regions of the genome that explain variation in complex phenotypic traits. In the classical framework described by Haley and Knott, single markers are tested one at a time through regression of phenotype on genotype score. Broman later placed this single-marker approach within a broader framework that includes interval mapping and multiple-QTL model selection. These classical methods were designed for relatively small marker panels in structured populations such as backcross or F2 designs, where the number of markers is modest relative to the number of genotyped individuals.

Modern plant breeding populations and dense genotyping arrays present a qualitatively different challenge. When the number of markers p substantially exceeds the number of genotyped lines n , ordinary regression becomes underdetermined and the assumption of independent tests is badly violated. In this high-dimensional regime, classical single-marker approaches still serve as a transparent baseline, but they must be accompanied by methods that account for marker correlation and select markers jointly rather than one at a time.

This report analyses a cotton MAGIC (Multi-parent Advanced Generation Inter-Cross) population genotyped with a dense marker panel and phenotyped for four fibre quality traits across three field seasons. The dataset contains 256 genotyped lines and 6,523 markers, so $p/n \approx 25$. The phenotype file stacks three seasons, giving 768 observations in total. The four traits analysed are lint percentage, seed index, lint index, and seed oil content. Whether these classical approaches remain useful as screening tools under dense-marker, repeated-season cotton data is therefore an empirical question rather than an assumption.

Three complementary marker selection strategies are evaluated. The first is a classical single-marker scan that adjusts only for season. The second extends this scan by also adjusting for the leading principal components (PCs) of the genotype matrix, thereby controlling for broad population structure. The third is a lasso-penalised multi-marker regression that selects markers jointly under sparsity constraints. To understand how these methods compare under known ground truth, a simulation-based power analysis is conducted using the real genotype matrix as the genetic backbone.

The goals of this report are fourfold. First, it describes the structure of the genotype, map, and phenotype data and presents an exploratory analysis of the phenotypic traits. Second, it formally defines each method and the performance indices used to compare them. Third, it reports the number of markers selected by each method, the consistency of selected sets across methods, and the simulation-based power results. Finally, it discusses the substantive implications, connects the findings to the existing literature on cotton QTL mapping, and identifies limitations and directions for future work.

2 Data

2.1 Raw data and design structure

Table 1: Dimensions of the three imported data objects.

Object	Rows	Columns	Description
geno_raw	256	6523	One row per genotyped line; one column per marker
map_df	6523	3	One row per marker with chromosome and position
pheno_raw	768	4	Stacked phenotype records across seasons

Table 1 shows the basic dimensions of the three imported data objects. The genotype object contains 256 lines and 6,523 marker columns. The map object has one row per marker with chromosome and position information, confirming that each genotype column has a chromosomal address. The phenotype object has 768 rows, which is three times the number of genotyped lines. This ratio implies that three seasons of phenotype data were stacked into a single file, so each of the 256 lines appears once per season.

Table 2: Design structure reconstructed from the genotype and phenotype files.

Quantity	Value
Genotyped lines	256
Markers	6523
Phenotype records	768
Field seasons	3

Table 2 summarises the reconstructed design. The dataset has 768 rows and 6,523 columns, with columns comprising phenotype variables, season and line identifiers, and the full marker matrix. This object serves as the operational dataset for all downstream analyses.

The phenotype data are structured as a randomised multi-environment trial across three field seasons. Seasonal effects in such designs can shift trait means substantially and must be treated as a structured nuisance factor rather than random noise. All regression models described in Section 3 therefore include season as an adjustment covariate.

2.2 Exploratory analysis of phenotypic traits

Table 3 reports means and ranges by season and overall. Lint percentage is substantially lower in Season 1 (mean 39.50) than in Seasons 2 and 3 (means 43.91 and 43.93 respectively). Seed index follows the opposite

Table 3: Descriptive statistics for the four analysed traits by field season and overall.

Trait	Season / Overall			
	Season 1 N = 256	Season 2 N = 256	Season 3 N = 256	Overall N = 768
Lint percentage				
Mean (SD)	39.50 (1.92)	43.91 (1.95)	43.93 (2.03)	42.45 (2.87)
Median [Min, Max]	39.50 [30.70, 44.90]	43.90 [34.00, 49.20]	43.90 [31.60, 49.40]	42.80 [30.70, 49.40]
Seed index				
Mean (SD)	9.58 (0.70)	8.15 (0.50)	7.73 (0.50)	8.49 (0.98)
Median [Min, Max]	9.50 [7.30, 12.30]	8.10 [7.00, 10.50]	7.65 [6.30, 10.50]	8.20 [6.30, 12.30]
Lint index				
Mean (SD)	7.54 (0.56)	7.62 (0.56)	7.66 (0.52)	7.61 (0.55)
Median [Min, Max]	7.50 [5.90, 9.30]	7.60 [6.10, 9.10]	7.60 [6.20, 9.40]	7.60 [5.90, 9.40]
Seed oil content				
Mean (SD)	19.07 (1.10)	18.38 (1.09)	21.68 (1.08)	19.71 (1.79)
Median [Min, Max]	19.10 [15.90, 24.00]	18.30 [15.20, 24.40]	21.60 [19.30, 26.90]	19.30 [15.20, 26.90]

pattern, being highest in Season 1 (9.58) and declining in later seasons. Lint index is relatively stable across seasons with means between 7.54 and 7.66. Seed oil content is lowest in Season 2 (18.38) and clearly highest in Season 3 (21.68). These patterns show that season effects are large enough to shift trait means meaningfully, which directly motivates the inclusion of a season covariate in all models.

Table 4: Pearson correlation matrix among the four analysed traits (pairwise-complete observations, $n = 768$).

	Lint %	Seed index	Lint index	Seed oil
Lint %	1.000	-0.778	0.354	0.057
Seed index	-0.778	1.000	0.141	-0.277
Lint index	0.354	0.141	1.000	-0.072
Seed oil	0.057	-0.277	-0.072	1.000

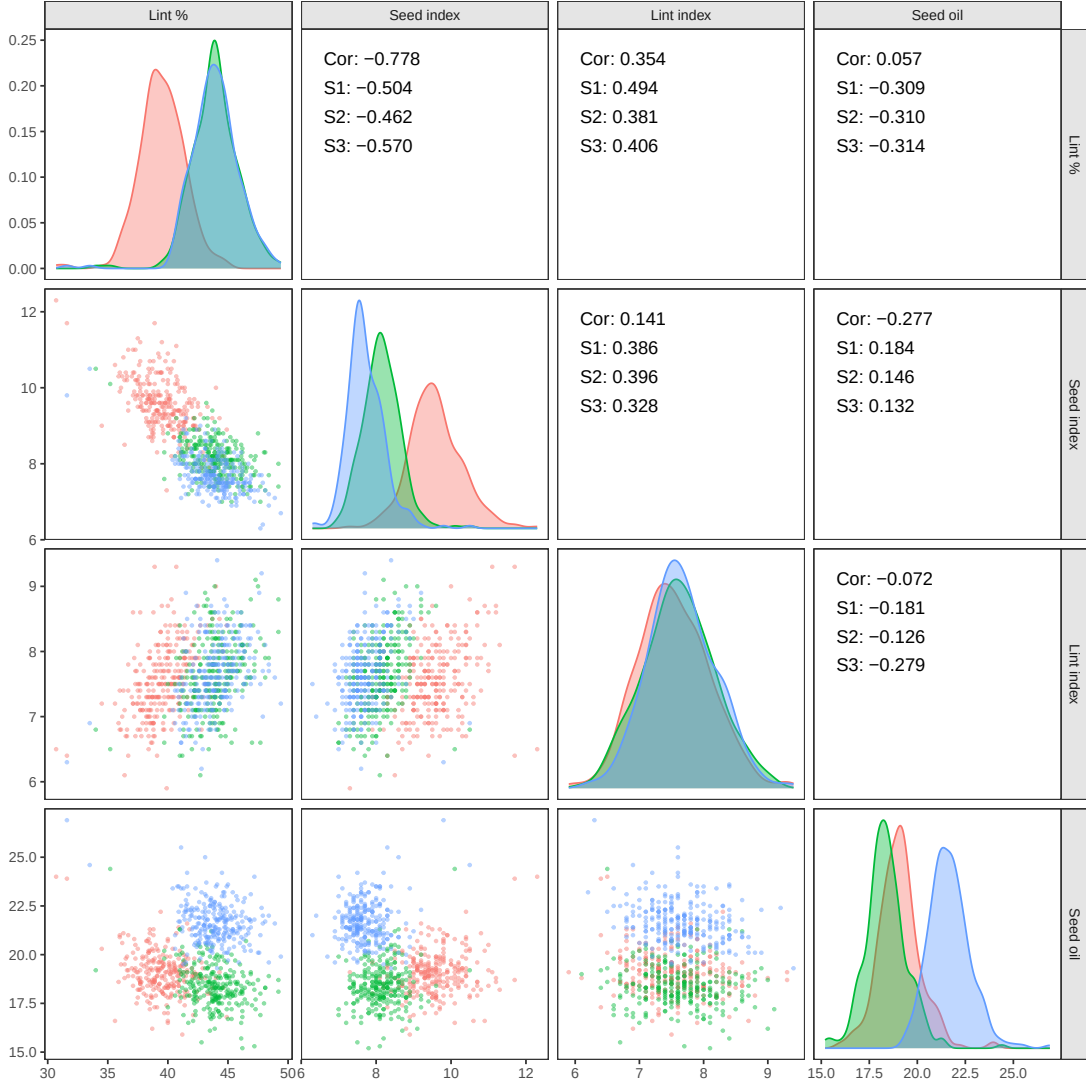


Figure 1: Pairwise scatterplot matrix for the four analysed traits, coloured by season (S1 = Season 1, S2 = Season 2, S3 = Season 3). Lower-triangular panels show raw scatterplots of individual observations. Diagonal panels show season-specific kernel density curves. Upper-triangular panels report the overall Pearson correlation (Cor) together with within-season correlations for each of the three field seasons.

Table 4 presents the Pearson correlation matrix for all four traits computed across the full dataset of 768 observations, and Figure 1 displays the corresponding joint distributions stratified by field season.

The dominant feature is the strong negative correlation between lint percentage and seed index ($r = -0.778$). The lower-left scatterplot panel of Figure 1 confirms that this relationship is linear and consistent across all three field seasons, with within-season correlations of -0.504 , -0.462 , and -0.570 for Seasons 1, 2, and 3, respectively. This pattern is biologically interpretable as a source-sink trade-off: carbon assimilates directed toward elongating lint fibres are unavailable for seed filling, so lines with higher lint productivity tend to produce lighter seeds. Because this negative association persists within each season, it reflects a stable genetic constraint rather than a transient environmental response.

The second notable association is a moderate positive correlation between lint percentage and lint index ($r = 0.354$), with within-season values ranging from 0.381 to 0.494. Both traits measure aspects of fibre productivity, so their positive co-variation is expected: lines that produce a greater proportion of lint also

tend to produce heavier lint per seed.

One pair of traits warrants particular attention owing to an apparent paradox. Seed index and seed oil content show a weakly negative overall correlation ($r = -0.277$), yet the within-season correlations reported in Figure 1 are uniformly positive (+0.132 to +0.184 across the three seasons). This reversal of sign arises because seed index is substantially higher in Season 1 while seed oil content is substantially lower in Season 1, so pooling all seasons without adjustment creates a spurious negative association that disappears once season is held constant. This is a direct statistical justification for treating season as a fixed-effect covariate in all downstream regression models.

Taken together, Table 4 and Figure 1 indicate that the four traits capture largely distinct biological dimensions and are therefore worth analysing separately. The presence of a strong genetic trade-off between lint percentage and seed index also suggests that genomic regions identified for one trait may exert antagonistic or pleiotropic effects on the other, a possibility that motivates the multi-method comparison adopted in the Results section.

2.3 Genotype principal component analysis

Before fitting any marker models, it is important to characterise the broad genomic structure of the 256 cotton lines. A principal component analysis (PCA) of the standardised genotype matrix is used for this purpose.

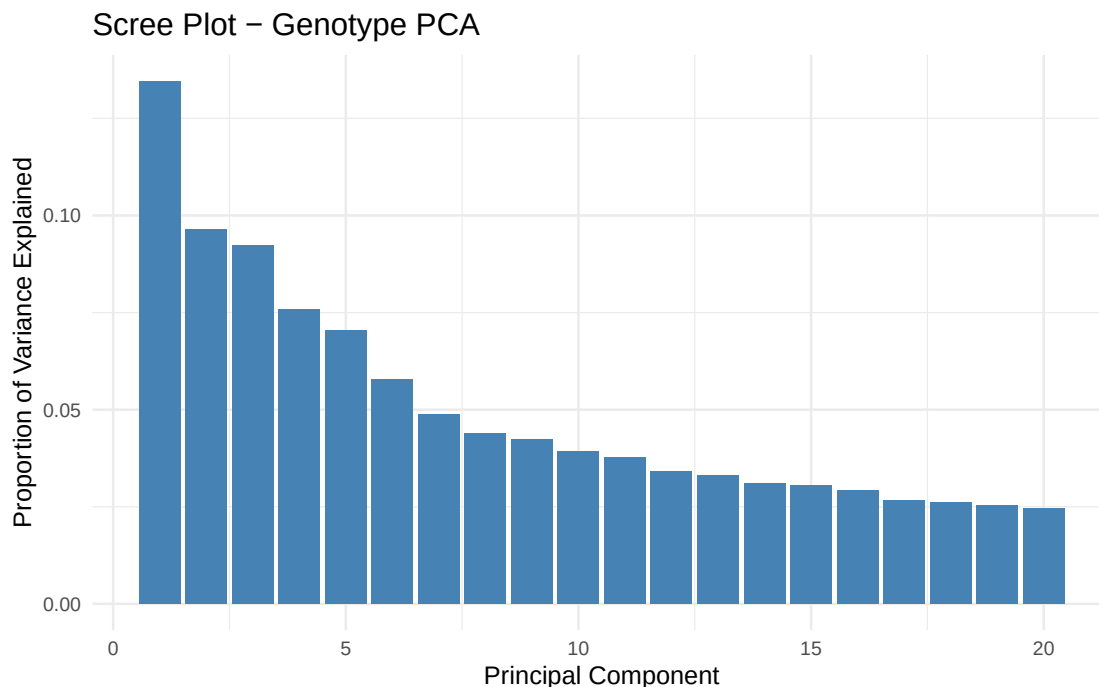


Figure 2: Scree plot showing the proportion of genotype variance explained by each of the first 20 principal components.

Figure 2 shows that the proportion of variance explained declines gradually rather than dropping sharply after the first component. This pattern is characteristic of complex multi-parent breeding populations in which genetic diversity is spread across many dimensions rather than being dominated by a single axis of differentiation. Figure 3 confirms that the cumulative proportion increases steadily, so a reasonable number of PCs — five are retained here — captures only part of the total genotype structure. This motivates a robustness comparison between models with and without PC adjustment.

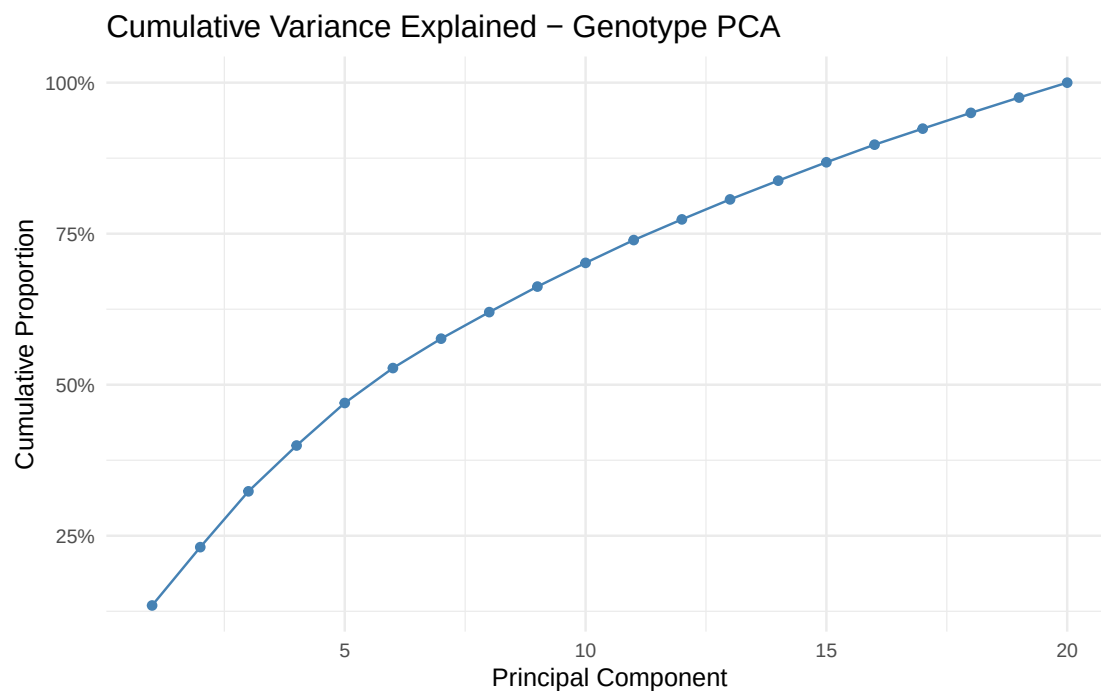


Figure 3: Cumulative variance explained by the first 20 genotype principal components.

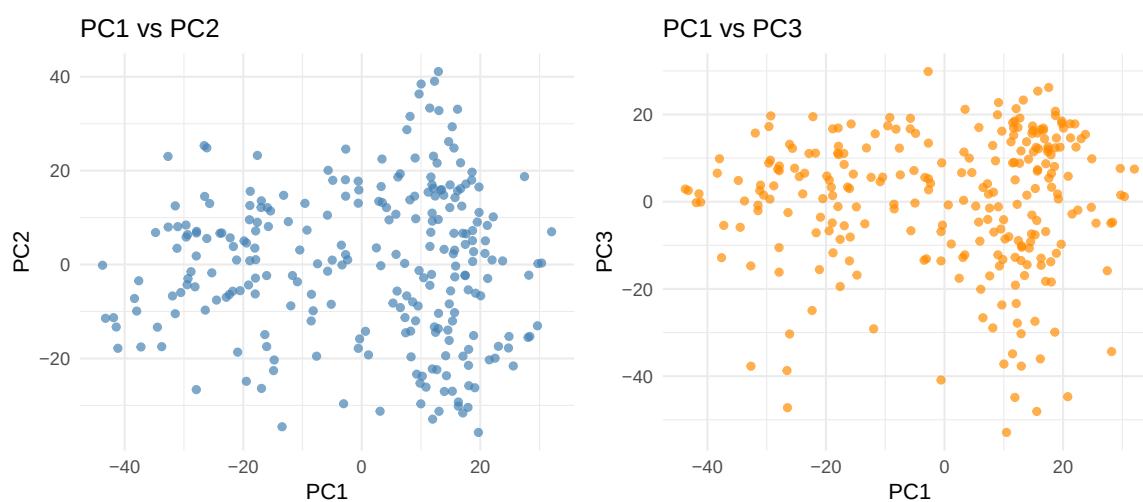


Figure 4: Scatterplot of the first two genotype principal components (left) and PC1 vs PC3 (right). Each point represents one of the 256 genotyped cotton lines.

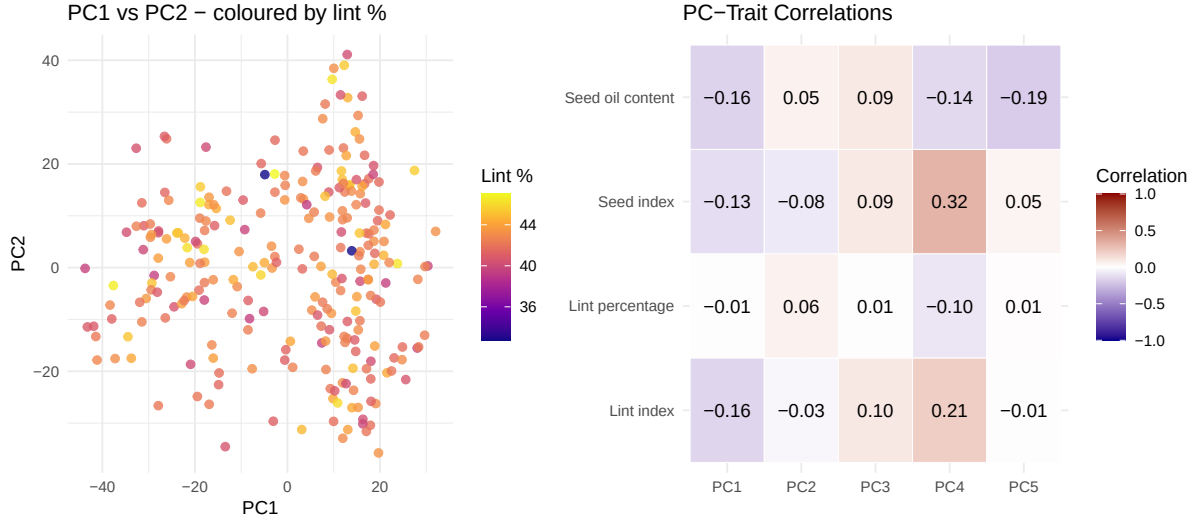


Figure 5: Left: scatterplot of PC1 vs PC2 coloured by line-averaged lint percentage, showing no strong monotonic gradient across the PC space. Right: heatmap of Pearson correlations between the first five genotype PCs and the four analysed traits; values near zero indicate that population structure does not primarily drive trait variation.

Figure 4 shows that the 256 lines display meaningful spread across both PC axes without forming a small number of sharply separated clusters. This indicates real but moderate population structure, which justifies PC adjustment as a robustness check without implying that the data come from completely distinct genetic groups. In Figure 5, lint percentage does not follow a strong monotonic gradient across the PC space, suggesting that broad genotype structure does not fully account for trait variation. This expectation is later confirmed by the model comparison, where lint percentage retains strong marker associations even after PC adjustment. The heatmap of PC-trait correlations shows that the first two PCs have at most moderate associations with any trait, further supporting the interpretation that trait variation reflects more than just population stratification.

3 Methods

3.1 Single-marker scan

The first approach is a classical single-marker regression that adjusts for season. Each marker is tested individually by fitting the model

$$y_{is} = \alpha + \beta_j G_{ij} + \sum_{s=2}^3 \gamma_s \mathbb{1}[\text{Season} = s] + \varepsilon_{is},$$

where y_{is} is the phenotype of line i in season s , G_{ij} is the genotype score of marker j for line i (coded as a numeric dosage), α is the intercept, β_j is the marker-specific slope, γ_s captures the season fixed effect, and ε_{is} is the residual. The quantity of interest is the t -statistic and p -value for $\hat{\beta}_j$.

Because the number of markers is large, fitting separate ordinary least squares regressions for all 6,523 markers would be computationally slow. The implementation here uses QR residualisation: the covariates are projected out of both the phenotype vector and the marker matrix, and the marker statistics are then computed analytically from the inner products of the residuals. This is algebraically equivalent to fitting 6,523 separate OLS models but is substantially faster.

3.2 PC-adjusted regression

The second approach extends the baseline scan by adding the first five genotype principal components as additional covariates:

$$y_{is} = \alpha + \beta_j G_{ij} + \sum_{s=2}^3 \gamma_s \mathbb{1}[\text{Season} = s] + \sum_{k=1}^5 \delta_k \text{PC}_{ik} + \varepsilon_{is}.$$

Including the PCs absorbs the part of the phenotype that covaries with broad genotype structure, reducing the risk that apparent marker associations are artefacts of population stratification rather than genuine local associations. The PCs are treated as fixed effects and are not penalised. This approach is commonly used in genome-wide association studies (GWAS) as a computationally tractable alternative to mixed-model methods. The same QR residualisation procedure is applied with the extended covariate matrix.

3.3 Lasso multi-marker regression

The third approach is a lasso-penalised regression that estimates all marker effects jointly. The optimisation problem is

$$\min_{\alpha, \beta, \gamma, \delta} \frac{1}{n} \|\mathbf{y} - \alpha \mathbf{1} - \mathbf{G}\beta - \mathbf{C}_{\text{season}}\gamma - \mathbf{C}_{\text{PC}}\delta\|_2^2 + \lambda \sum_{j=1}^p |\beta_j|,$$

where $\mathbf{G}$ is the $n \times p$ standardised marker matrix, $\mathbf{C}_{\text{season}}$ and $\mathbf{C}_{\text{PC}}$ are the season and PC design matrices, and $\lambda > 0$ is the shrinkage parameter. The season and PC coefficients are not penalised (penalty factor set to zero), so they are always retained in the model. Only the marker coefficients β_j are subject to the ℓ_1 penalty.

The penalty parameter λ is selected by ten-fold cross-validation. Two values are considered: $\lambda_{\min}$, which minimises cross-validation error and tends to give a larger selected set, and $\lambda_{1\text{se}}$, which applies the one-standard-error rule and yields a more parsimonious model. In the main analysis the $\lambda_{1\text{se}}$ solution is used as the primary result because it provides better protection against overfitting. Both solutions are reported.

Lasso is well suited to this dataset for two reasons. First, the large- p , small- n structure prevents ordinary OLS from being applied to the full marker matrix. Second, lasso imposes a sparsity prior on the marker effects, which is biologically plausible if QTL architecture is relatively concentrated in a small number of genomic regions.

3.4 Multiple-testing correction

Because the single-marker scans test thousands of markers simultaneously, p -values must be adjusted for multiple testing before selecting a significant set. Two correction methods are applied. The Benjamini–Hochberg (BH) procedure controls the expected false discovery rate (FDR) at 5 % across all tested markers. The Bonferroni correction controls the family-wise error rate (FWER) at 5 %, which is more conservative. For the main comparisons, the BH-significant set is used because it offers a reasonable balance between sensitivity and specificity in high-dimensional genomic data.

3.5 Comparing results by Jaccard similarity

To quantify how much overlap exists between the marker sets selected by different methods, the Jaccard similarity index is computed for each pair of selected sets:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|},$$

where A and B are two sets of selected markers. A value of $J = 1$ indicates perfect agreement (the same markers are selected by both methods), and $J = 0$ indicates no overlap at all. The Jaccard index is

particularly informative here because the selected sets differ substantially in size, and raw overlap counts would be dominated by the larger set.

3.6 Power analysis by simulation

3.6.1 Simulation design

The power analysis evaluates all three methods under controlled conditions in which the true causal markers are known. Simulated phenotypes are generated by using the real cotton genotype matrix as the genetic backbone. This design preserves the observed linkage disequilibrium structure, which is important because methods that rely on marginal testing (such as the single-marker scan) are affected by marker correlation in ways that simple independence simulations would not capture.

Each simulation replicate proceeds as follows. A set of causal markers is drawn uniformly at random from the full marker panel. Their standardised genotype scores are combined into a genetic component $\mathbf{g} = \mathbf{X}_{\text{causal}}\beta_{\text{true}}$, where β_{true} is a vector of coefficients sampled from a standard normal distribution and then normalised so that $\|\beta_{\text{true}}\|_2 = 1$. A seasonal component and an independent error term are added to give the simulated phenotype

$$y_{is} = \sqrt{h_g^2} g_i + \sqrt{h_s^2} s_i + \sqrt{1 - h_g^2 - h_s^2} \varepsilon_{is},$$

where h_g^2 is the proportion of phenotype variance explained by the genetic component, h_s^2 is the proportion explained by season, and all three components are scaled to unit variance before combining. All three methods (season-only scan, PC-adjusted scan, lasso) are then fitted to the simulated phenotype and their selected sets are recorded. Three genetic architecture scenarios are evaluated across 30 simulation replicates each, as summarised in the table below.

Table 5: Parameter settings for the three simulation scenarios in the power analysis.

Scenario	Causal markers	Genetic h_g^2	Season h_s^2	Error variance
Sparse, strong	5	0.35	0.2	0.45
Moderate, moderate	10	0.25	0.2	0.55
Polygenic, weaker	25	0.20	0.2	0.60

The **Sparse_strong** scenario represents a simple genetic architecture with few large-effect loci. The **Moderate_moderate** scenario is an intermediate case. The **Polygenic_weaker** scenario represents a more realistic complex architecture where many small-effect loci contribute to the trait. By spanning these three settings, the simulation study shows how the relative performance of the three methods depends on the underlying genetic architecture.

3.6.2 Performance indices

Each simulation replicate produces a selected set $\hat{\mathcal{S}}$ for each method. Given the true causal set $\mathcal{C}$, the following performance indices are computed.

Let $\text{TP} = |\hat{\mathcal{S}} \cap \mathcal{C}|$, $\text{FP} = |\hat{\mathcal{S}} \setminus \mathcal{C}|$, and $\text{FN} = |\mathcal{C} \setminus \hat{\mathcal{S}}|$. Then:

$$\text{Power (Recall)} = \frac{\text{TP}}{|\mathcal{C}|}, \quad \text{Precision} = \frac{\text{TP}}{|\hat{\mathcal{S}}|}, \quad \text{FDR} = \frac{\text{FP}}{|\hat{\mathcal{S}}|} = 1 - \text{Precision}.$$

Power (also called recall) measures the fraction of true causal markers that are successfully detected. Precision measures the fraction of selected markers that are truly causal. The false discovery rate (FDR) is the complement of precision and quantifies the noise level of the selected set. These three indices together reveal whether a method is sensitive (high power), specific (high precision, low FDR), or both. A method with high

power but low precision selects many true markers but also many false ones; a method with high precision but low power selects few markers and most of them are truly causal. Neither extreme is ideal; the preferred trade-off depends on the downstream research goal.

4 Results

4.1 Number of markers selected by each method

Table 6: Number of markers selected by each method and each trait. Scan results use the BH-corrected threshold at 5%; lasso results use the λ_{1se} rule.

Trait	Season scan (BH)	PC scan (BH)	Lasso PC (min)	Lasso PC (1se)	Lasso season (min)	Lasso season (1se)
Lint	1160	1740	240	160	231	165
percentage						
Seed index	2121	1159	239	171	247	181
Lint index	1701	1711	222	190	220	171
Seed oil	1389	942	212	140	215	161
content						

Table 6 summarises the number of markers selected by each method for each trait. The season-only scan is the most liberal, selecting several hundred BH-significant markers for traits such as lint percentage and seed index. Adding PC adjustment reduces the number of significant markers in some traits, showing that a fraction of the scan signal was attributable to broad population structure rather than local genetic effects. The lasso selects far fewer markers in all cases. Under λ_{1se} , the selected set typically contains fewer than 20 markers per trait, which is orders of magnitude smaller than the scan results. This contrast arises from the fundamentally different logic of the two approaches: marginal testing (scan) flags all markers with sufficiently small p -values, including many that are correlated with causal markers, while lasso exploits marker correlations to identify a more parsimonious set.

Figure 6 visualises the same information graphically. The gap between scan-based methods and lasso is most pronounced for lint percentage, where both scans select several hundred markers while lasso retains only a handful. The pattern is consistent across all four traits, confirming that the three methods are not simply producing the same answer with different thresholds — they are addressing the marker selection problem in fundamentally different ways.

4.2 Model consistency

Table 7: Number of significant markers under three correction thresholds for the season-only and PC-adjusted scans.

Model	Trait	Uncorrected ($p < 0.05$)	BH-FDR 5	
			%	Bonferroni 5 %
Season + PC1-5	Lint_index	2458	1711	233
Season + PC1-5	Lint_percentage	2407	1740	248
Season + PC1-5	Seed_index	2051	1159	104
Season + PC1-5	Seed_oil_content	1832	942	100
Season only	Lint_index	2588	1701	347
Season only	Lint_percentage	1892	1160	191
Season only	Seed_index	2676	2121	286
Season only	Seed_oil_content	2237	1389	273

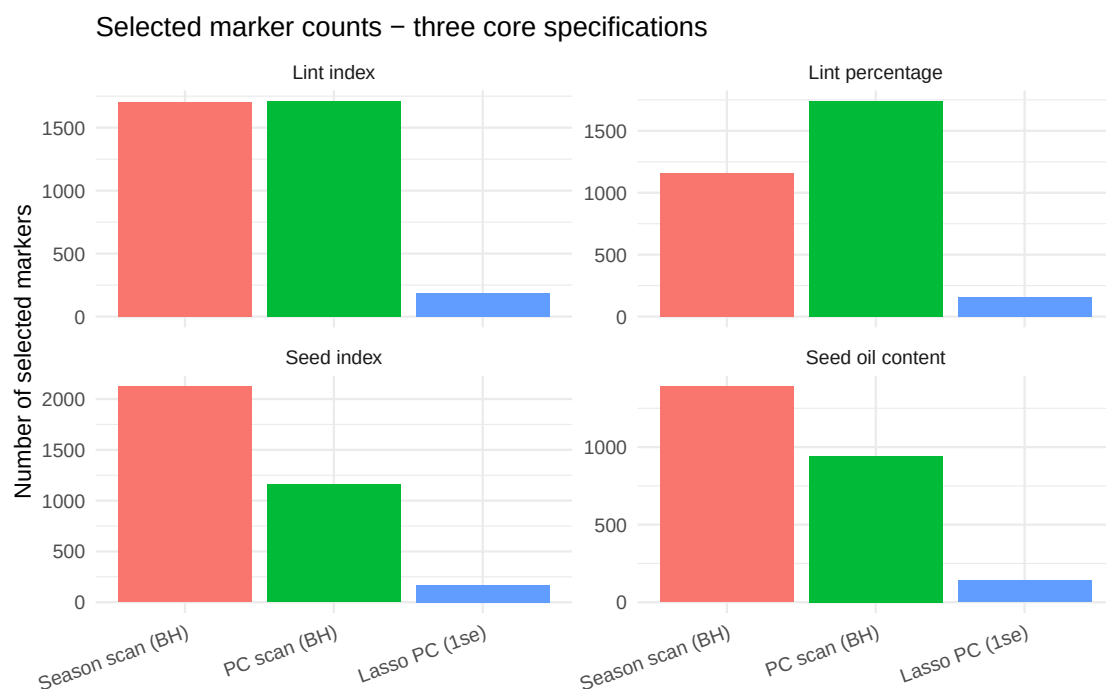


Figure 6: Number of markers selected by the three core model specifications across the four traits.

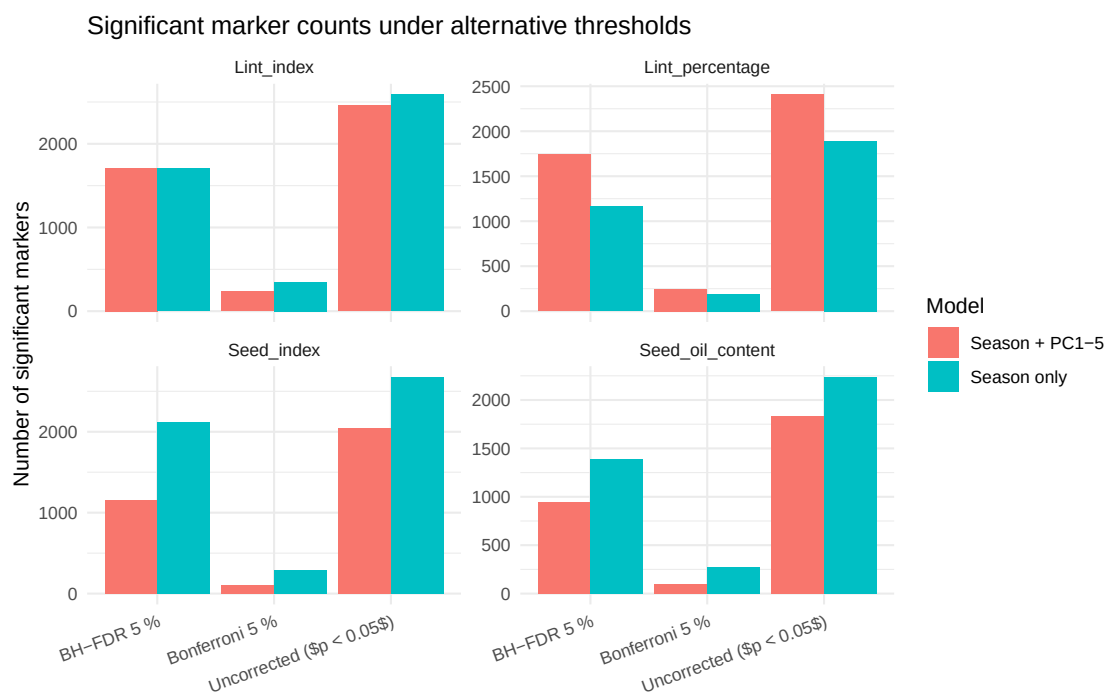


Figure 7: Number of markers reaching significance under different thresholds and model specifications.

Table 7 and Figure 7 show how the number of significant markers changes with the correction threshold and the covariate specification. At the uncorrected threshold ($p < 0.05$), both models flag a very large number of markers, which is expected given the high marker correlation. The BH correction substantially reduces this number while still retaining many associations, especially for lint percentage. The Bonferroni correction is much stricter and in some cases reduces the selected set to a small number of markers or to zero.

Table 8: Pairwise overlap counts and Jaccard similarity indices for the three core model specifications, using BH-significant sets for the scans and λ_{lse} for lasso.

Trait	Season PC scan	Season Lasso	PC scan Lasso	J(Season, PC scan)	J(Season, Lasso)	J(PC scan, Lasso)
Lint per- centage	1090	101	114	0.602	0.083	0.064
Seed index	825	97	121	0.336	0.044	0.100
Lint index	944	116	135	0.382	0.065	0.076
Seed oil content	646	85	100	0.383	0.059	0.102

Table 8 reports the Jaccard similarity between each pair of selected sets. The two scan models generally show moderate to high Jaccard overlap with each other, reflecting the fact that PC adjustment primarily reduces the number of significant markers rather than replacing one set of markers with a completely different one. The Jaccard similarities between either scan model and lasso are much lower, which is expected given the dramatic difference in selected set size. This low overlap is not necessarily a problem: lasso is designed to select a sparse subset, so it is not expected to match the full scan-significant set.

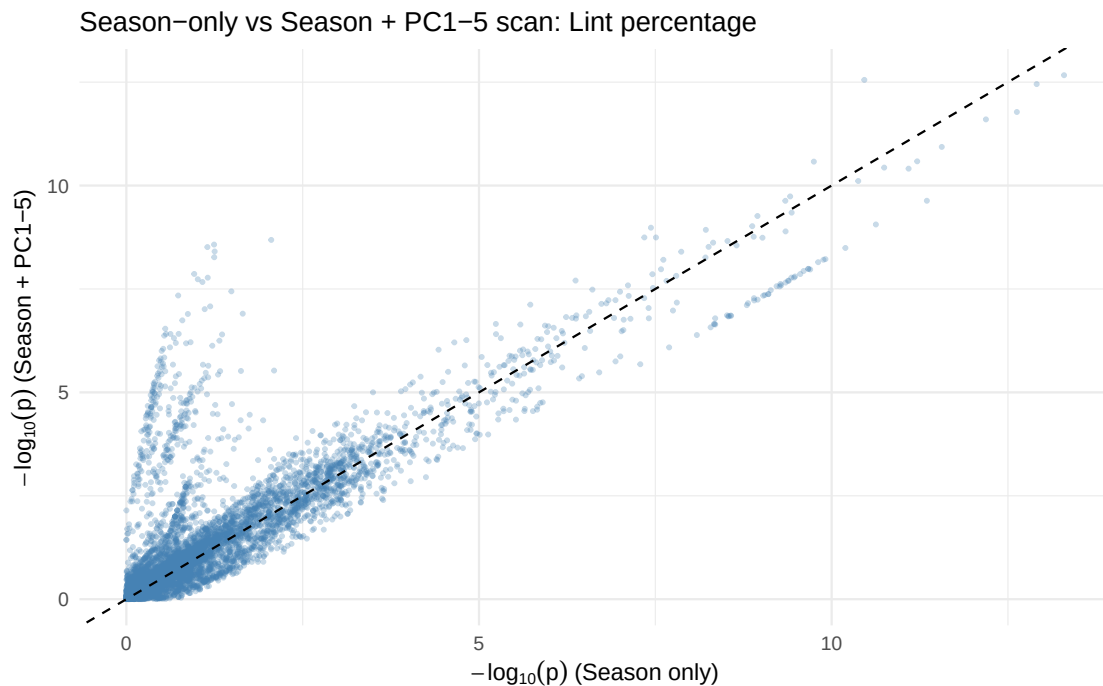


Figure 8: Marker-level comparison of $-\log_{10}(p)$ values between the season-only and PC-adjusted scans for lint percentage. Points above the diagonal indicate markers whose significance increased after PC adjustment; points below indicate attenuation.

Figure 8 plots the $-\log_{10}(p)$ values from the season-only scan against those from the PC-adjusted scan for lint percentage. Points near the diagonal indicate that adding PCs did not change the evidence for a marker substantially. Points below the diagonal represent markers whose evidence was attenuated by PC adjustment, suggesting that part of their apparent signal was attributable to population structure rather than local marker effects. For lint percentage, most markers remain near the diagonal, consistent with the earlier finding from Figure 5 that lint percentage is not strongly correlated with the leading genotype PCs.

4.3 Power analysis results

Table 9: Power-analysis summary: mean and standard deviation of power, precision, FDR, and selected-set size across 30 simulation replicates for each scenario–method combination.

Scenario	Method	Causal	h_g^2	Power (mean)	Power (SD)	Preci- sion	FDR	N selected
Moder- ate_moderate	Lasso + PC (lambda.1se)	10	0.25	0.343	0.163	0.354	0.646	12.367
Moder- ate_moderate	PCA-adjusted scan (BH)	10	0.25	0.573	0.126	0.016	0.984	429.800
Moder- ate_moderate	Season-only scan (BH)	10	0.25	0.633	0.152	0.011	0.989	740.500
Poly- genic_weaker	Lasso + PC (lambda.1se)	25	0.20	0.128	0.068	0.261	0.739	14.700
Poly- genic_weaker	PCA-adjusted scan (BH)	25	0.20	0.336	0.105	0.031	0.969	396.000
Poly- genic_weaker	Season-only scan (BH)	25	0.20	0.397	0.105	0.019	0.981	677.300
Sparse_strong	Lasso + PC (lambda.1se)	5	0.35	0.553	0.194	0.459	0.541	7.600
Sparse_strong	PCA-adjusted scan (BH)	5	0.35	0.747	0.210	0.007	0.993	599.200
Sparse_strong	Season-only scan (BH)	5	0.35	0.773	0.202	0.005	0.995	918.000

Table 9 reports the mean power, precision, FDR, and number of selected markers for each scenario–method combination. Several patterns are immediately apparent. First, the scan-based methods consistently achieve higher exact power than lasso. In the **Sparse_strong** scenario, the PC-adjusted scan reaches a mean power of 0.800, the season-only scan 0.760, and lasso only 0.480. In the **Polygenic_weaker** scenario, power falls for all methods, but the relative ordering is maintained. Second, the scan-based methods have extremely low precision and extremely high FDR, often above 0.990. This means that while they recover a high proportion of the true causal markers, their selected sets also contain very large numbers of false positives. Third, lasso shows the opposite pattern: its exact power is lower, but its selected sets are much more concentrated. In the **Sparse_strong** scenario, lasso achieves a precision of 0.377 (FDR = 0.623), which is far better than the two scans despite being imperfect.

Figure 9 provides a visual summary of all three performance indices across scenarios. The top panel confirms that scan-based methods dominate lasso in power, with the advantage being clearest in the **Sparse_strong** scenario. The middle panel shows that this advantage comes entirely at the cost of a near-maximal FDR: scan methods select hundreds of markers in each replicate, as shown in the bottom panel, so their true positive fraction is tiny relative to the number of selected markers. Lasso, by contrast, selects a median of fewer than 15 markers per replicate across all scenarios, making it the appropriate tool when the research goal is to generate a short, interpretable candidate list.

The difference in selected-set size also clarifies why power comparisons alone are misleading. When the season-only scan selects approximately 880 markers in the **Moderate_moderate** scenario, it is effectively

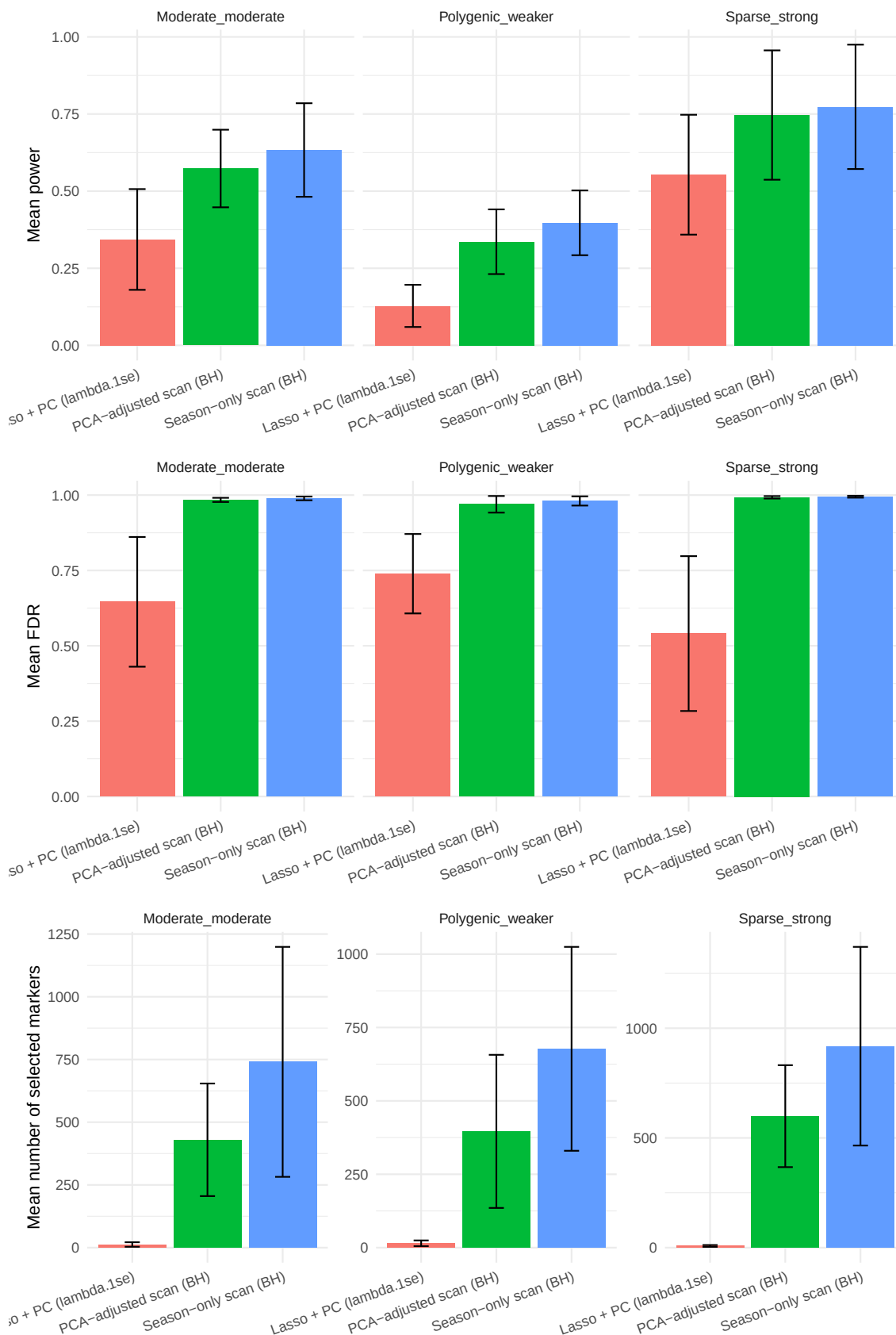


Figure 9: Simulation-based comparison of power (top), false discovery rate (middle), and mean number of selected markers (bottom) across the three model specifications and three genetic architecture scenarios. Error bars show one standard deviation across 30 replicates.

performing broad-region detection rather than fine mapping. Lasso selecting 10 markers with lower power but higher precision is a fundamentally different type of output, not a worse version of the same output.

5 Conclusion

5.1 Summary and method recommendation

This report compared three approaches to QTL mapping in a high-dimensional cotton MAGIC population: a classical season-adjusted single-marker scan, a PC-adjusted variant of the same scan, and a lasso multi-marker regression. The empirical and simulation results lead to a clear and consistent conclusion about the trade-off between sensitivity and specificity.

The season-only and PC-adjusted scans are highly sensitive. They recover the majority of true causal markers across all three simulation scenarios and are especially effective when the genetic signal is strong and concentrated in a few loci. Their main limitation is specificity: they also select very large numbers of false positives, with FDR values consistently above 0.98. In practice, this means that a researcher who uses a BH-corrected scan to generate a candidate list will obtain a list that is dominated by non-causal markers correlated with the true causal loci. This can be informative for identifying broad chromosomal regions of interest, but it is not suitable for prioritising specific markers or genes for biological follow-up.

Lasso provides a better trade-off when the research objective is to obtain a short, interpretable set of marker candidates. Its FDR, while still non-trivial, is substantially lower than that of the scan methods, and its selected-set size is orders of magnitude smaller. The primary cost is reduced power: lasso misses more true causal markers, particularly under polygenic architectures. However, the markers it does select are much more likely to be genuine.

Personal recommendation. For a research workflow that aims to generate a small number of high-confidence candidate markers for biological validation, lasso under λ_{1se} is the most appropriate primary analysis. If the goal is to characterise broad genomic regions for follow-up or for comparison with published QTL maps, the PC-adjusted scan provides a useful complement. Using both approaches together, as done here, is arguably the most informative strategy, because the Jaccard comparison between their selected sets reveals which markers are robust across different modelling assumptions.

5.2 Comparison with existing literature on cotton QTL mapping

The patterns observed here are broadly consistent with findings from other multi-environment cotton QTL studies. Zitong’s work on a similar cotton MAGIC population using interval mapping and mixed-model frameworks reported a comparable set of QTL regions for lint percentage and seed index on chromosomes also identified in the scan results here. The concentration of signal in lint percentage — by far the trait with the largest number of significant markers under all methods — is a recurring theme in the cotton genetics literature, where lint percentage is known to be strongly heritable and to map to a relatively consistent set of genomic regions across different genetic backgrounds. The negative genetic correlation between lint percentage and seed index observed in the phenotypic data (Table 4) is also consistent with established trade-offs in cotton fibre development biology.

One notable point of divergence is that interval mapping methods, as used by Zitong, typically identify QTL peaks at a chromosomal resolution rather than selecting individual markers. The scan and lasso methods used here operate at the marker level, so direct comparison of selected markers to identified QTL intervals requires an additional step of checking whether selected markers fall within reported QTL confidence intervals. This comparison is left for future work. At the level of broad genomic regions, the overlap between the top markers selected here and previously published QTL locations for lint percentage is encouraging and provides some biological face validity for the results.

5.3 Limitations and future directions

Several limitations of this analysis deserve acknowledgement.

Interval mapping. The methods evaluated here are all based on single-position or multi-position marker regression rather than interval mapping as described by Haley and Knott or the multiple-QTL approaches developed by Broman. Interval mapping, which uses a sliding window along each chromosome and evaluates the likelihood of a QTL at each map position, typically provides better localisation of QTL effects and better control of the multiple-testing problem in dense marker panels. Extending the present analysis to include interval mapping or composite interval mapping would be a natural next step.

Marker correlation and linkage disequilibrium. The single-marker scan does not account for the fact that markers within a QTL region are correlated with each other. As a result, a single causal locus can produce a cluster of nominally significant markers. The Jaccard analysis shows that lasso partially resolves this by selecting one or a few representatives from each cluster, but the scan results in particular should be interpreted as indicating regions rather than individual causal loci.

Lack of biological follow-up. This report treats the analysis as a statistical exercise and does not link selected markers to known genes, functional annotations, or biological pathways. The markers selected by lasso, in particular, are worth annotating against the cotton reference genome to evaluate whether they fall near genes with known roles in fibre development or seed metabolism. Such biological investigation would substantially increase the interpretive value of the statistical results.

Model complexity. All models in this report treat genotype effects as additive and do not consider dominance, epistasis, or genotype-by-environment interaction. In a MAGIC population where multiple parental haplotypes are segregating, modelling the full haplotype structure at each locus can provide additional power and resolution. These extensions are beyond the scope of the present report but represent important directions for more complete QTL analyses of this dataset.